

Multi-Source Candidate Data Transformer

A Deterministic Trust-Based Candidate Profile Consolidation Engine — Design Document

Abstract. Modern recruitment platforms ingest candidate information from heterogeneous, often conflicting sources — recruiter CSVs, ATS exports, resumes, GitHub profiles, and recruiter notes. This project addresses the underlying challenge of **entity resolution** and **data fusion** rather than simple parsing: mapping disparate schemas into one canonical model, normalizing values, resolving conflicts, and computing per-field confidence so that downstream AI hiring systems can trust the output. The system follows a deterministic design philosophy — **"wrong but confident is worse than honestly empty"** — returning `null` rather than guessing when evidence is insufficient. The engine parses multiple structured and unstructured sources through an extensible parser registry, canonicalizes and normalizes field values, performs entity matching across sources, resolves conflicts using a trust-weighted scoring model inspired by linked-data fusion literature, and emits a fully provenance-tracked, configurably-projected candidate JSON record. The pipeline is orchestrated by a single entry point and exposed both as a CLI and as a REST API with a lightweight web front end for interactive use.

1. Problem & Objective

Downstream hiring AI requires a single clean profile per candidate. Incorrect merged data can silently bias decisions. The objective is to build a deterministic engine that consolidates structured and unstructured candidate data into one canonical profile while preserving **traceability**, **confidence**, **robustness**, and **configurability**, without ever inventing missing information.

2. Pipeline Overview

Sources → Parser Registry → Schema Mapping → Extraction → Canonicalization → Normalization → Validation → Entity Matching → Conflict Resolution → Trust Engine → Golden Record → Projection → Final JSON

3. Input Sources & Reliability

Source	Type	Key Contents	Reliability
Recruiter CSV	Structured	Name, Email, Phone, Company, Title	0.90
ATS JSON	Structured	Nested experience, education, certifications	0.95
Resume (PDF)	Unstructured	Experience, projects, skills, education	0.70
Recruiter Notes	Unstructured	Observations, interview notes	0.50
GitHub API	Semi-structured	Name, bio, repos, languages	0.85

4. Canonical Schema

```
candidate_id · full_name · emails[] · phones[] · links · headline · years_experience · skills[] · experience[] · education[] · provenance[] · overall_confidence
```

5. Architecture Components

- Parser Registry** `parsers/registry.py` — routes to independent parsers: `csv_parser`, `ats_parser`, `resume_parser`, `notes_parser`, `github_parser`; each emits a uniform Candidate Object.
- Schema Mapping** `mapper/schema_mapper.py` — translates source-specific field names (e.g. `candidateName`) into canonical fields (`full_name`).
- Extraction** `extractor/extractor.py` + `skills_vocab.py` — pulls structured signals (email, phone, experience, education, skills) from each source.
- Canonicalization** `canonicalizer/canonicalizer.py` — collapses synonymous values (e.g. "Google LLC" / "Google Inc." → "Google"; "CPP" → "C++").
- Normalization** `normalizer/normalizer.py` — phones → E.164, dates → YYYY-MM, country → ISO code, emails → lowercase.
- Validation** `validator/validator.py` + `schema_validator.py` — Pydantic-based checks; malformed input returns `null`, never crashes.

6. Entity Matching

`matcher/matcher.py` determines whether records across sources belong to the same person, using a priority cascade:

Email → Phone → GitHub → LinkedIn → `RapidFuzz(Name)` → Company

7. Conflict Resolution & Trust Engine

When sources disagree (e.g. "Google LLC" vs "Google" vs "Google Inc."), `trust/trust.py` — the **Trust Engine** — computes per-field confidence as a function of source reliability, cross-source agreement, and degree of conflict (with conflict handling in `resolver/resolver.py`), inspired by *Linked Data Integration with Conflicts* (Michelfeit et al., 2014). Values below a confidence threshold are returned as `null` instead of being guessed, directly enforcing the "honestly empty over wrong but confident" principle.

8. Merge Policy

Field	Strategy
Name	Highest confidence wins
Skills	Union across sources
Experience	Merged / deduplicated
Emails / Phones	Unique-set union

9. Provenance

`provenance/provenance.py` retains, per output field: source, parser, normalization method, matching reasons, and confidence score — enabling full explainability of the golden record (`schemas/canonical.py`).

10. Projection Layer & Output

`projection/projector.py` maps: internal canonical record → runtime config (`config/settings.py`, `config/sample_config.json`) — field selection, renaming, normalization rules, missing-value policy, confidence/provenance toggles → validated default or custom-schema JSON output, with no core engine modification required.

11. Technology Stack & Interfaces

Python 3.12 | `pdfplumber` | `pandas` | `phonenumbers` | `dateparser` | `pycountry` | `RapidFuzz` | `Pydantic` | `Requests` / GitHub REST API | `argparse` (main.py) | `pytest` | REST API (api.py) | Frontend: HTML/CSS/JS

Orchestrated end-to-end by `main_pipeline.py`, runnable via CLI (`main.py`) or as a REST service (`api.py`) with a lightweight `frontend/` UI for interactive uploads.

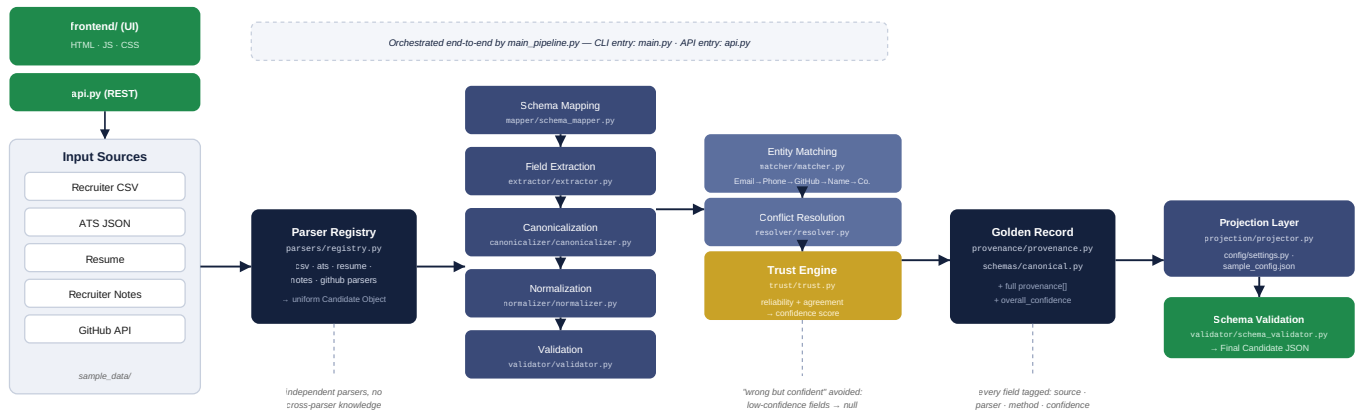
12. Key Assumptions & Limitations

- Email and phone are the strongest identity signals; unknowns are never inferred.
- Confidence scoring is fully deterministic; reliability weights are manually configured.
- Rule-based resume extraction (not LLM-based); no LinkedIn scraping; canonical dictionaries require maintenance.

13. References

- Michelfeit, Knap & Nečaský (2014). *Linked Data Integration with Conflicts* (ODCS-FusionTool) — trust scoring & resolution.
- Li et al. (2016). *A Survey on Truth Discovery* — source reliability estimation.
- Li et al. (2014). *Resolving Conflicts in Heterogeneous Data by Truth Discovery and Source Reliability Estimation (CRH)* — evidence weighting.

14. Architecture Diagram



Summary: A deterministic, explainable, trust-based data fusion engine that transforms heterogeneous candidate information from multiple structured and unstructured sources into a single canonical profile — with normalized fields, provenance tracking, confidence estimation, configurable output projection, and robust conflict resolution — ensuring uncertain information is returned as `null` rather than incorrectly inferred.